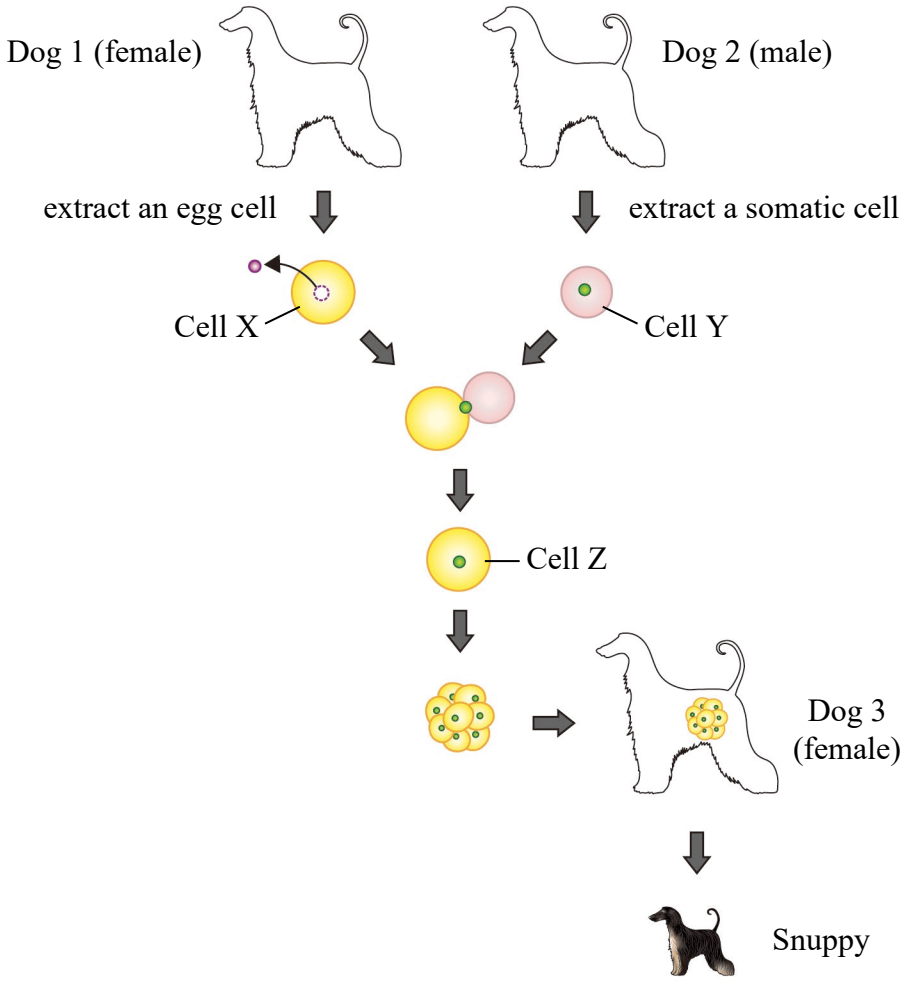
A. Multiple-choice questions

|!|BCEMB0410000035|!| ★

Directions: The following three questions refer to the diagram below, which shows the steps in the cloning of Snuppy, an Afghan hound.



Which of the following combinations is correct?

	Dog 1	***Dog 2***	***Dog 3***
A.	Egg donor	Nucleus donor	Surrogate mother
B.	Egg donor	Surrogate mother	Nucleus donor
C.	Nucleus donor	Egg donor	Surrogate mother
D.	Nucleus donor	Surrogate mother	Egg donor

##A##

Skill assessed	Mark
Applying concepts	1

© 2024 Aristo Educational Press Ltd.

|!|BCEMB0410000036|!| ★

Which of the following statements about Snuppy is/are correct?

(1) Snuppy is male.
(2) Snuppy is genetically identical to Dog 1.
(3) Snuppy looks exactly the same as Dog 2.

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (1), (2) and (3)

##A##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0410000037|!| ★

Which of the cells in the above diagram are diploid?

A. X and Y only
B. X and Z only
C. Y and Z only
D. X, Y and Z

##C##

Skill assessed	Mark
Applying concepts	1

© 2024 Aristo Educational Press Ltd.

|!|BCEMB0420000022|!| ★

Directions: The following two questions refer to the information below.

Cystic fibrosis is a genetic disease caused by a defective gene. The normal gene codes for the cystic fibrosis transmembrane protein (CFTP). Owing to the defective gene, cystic fibrosis patients lack the CFTP and suffer from thick mucus, which builds up in the respiratory tract. Gene therapy may offer a treatment for cystic fibrosis patients. Scientists are studying the usage of liposomes and viruses to deliver the normal gene into the cells lining the respiratory tract.

Which of the following comparisons between liposomes and viruses as vectors for gene therapy for cystic fibrosis is ***incorrect***?

	Liposomes	***Viruses***
A.	enclose DNA	enclose only RNA
B.	do not cause diseases	can be modified to become harmless
C.	fuse with cell membranes so that the gene enclosed can enter cells	attach on cell surfaces and injects its genetic material into cells
D.	relatively safe but less effective	may become pathogenic again but more effective

##A##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0420000023|!| ★

In this example of gene therapy, the treatment is successful if

A. the vectors enter the respiratory tract.
B. the normal gene is inserted into the DNA of the cells lining the respiratory tract.
C. the normal gene is expressed in the the cells lining the respiratory tract.
D. the vectors do not cause side effects, such as allergic reaction.

##C##

Skill assessed	Mark
Applying concepts	1

© 2024 Aristo Educational Press Ltd.

|!|BCEMB0420000015|!| ★

Directions: The following three questions refer to the information below.

Corn borers are insects that feed on corn plants. Scientists can modify the genome of corn plants so that the genetically modified corn plants product Bt toxin which kills corn borers.

Which of the following statements about the genetically modified corn plants described above is ***incorrect***?

A. The gene for Bt toxin could be transferred into corn plant cells using *Agrobacterium*.
B. The Bt toxin produced by the GM corn plants kills any animals that feed on the plants.
C. Insects which have resistance against Bt toxin can eat the GM corn plants and still survive and reproduce.
D. Farmers can use less pesticide on these GM corn plants.

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0420000016|!| ★

Which of the following is/are the advantage(s) of planting these GM corn plants?

(1) Biodiversity is increased.
(2) The growth of weeds would be retarded.
(3) Chemical pollution due to the heavy use of pesticides can be reduced.

A. (1) only
B. (2) only
C. (3) only
D. (2) and (3) only

##C##

Skill assessed	Mark
Understanding basic concepts	1

© 2024 Aristo Educational Press Ltd.

|!|BCEMB0420000017|!| ★

One disadvantage of planting these GM corn plants is that

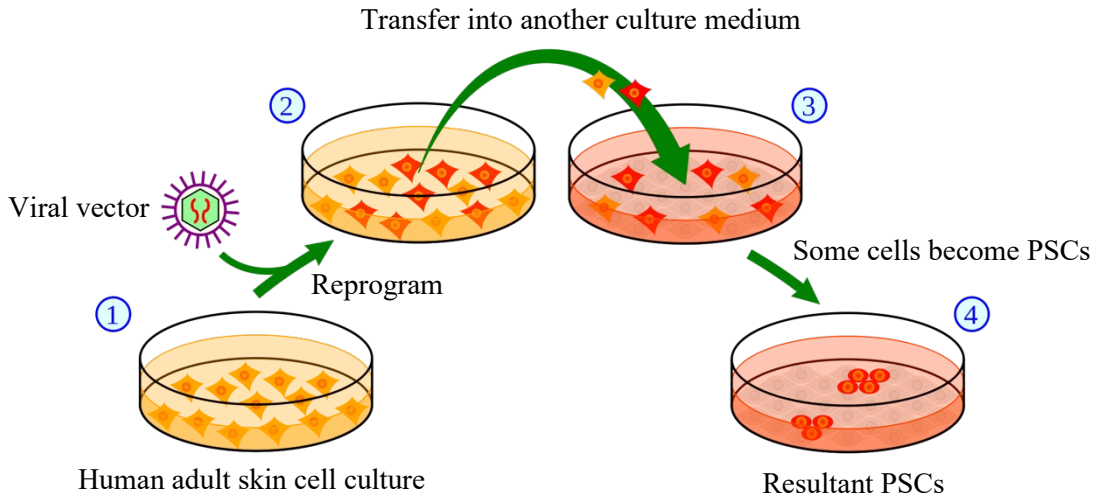
A. corn borers would become extinct.
B. the GM corn plants are not suitable for human consumption.
C. corn borers would acquire resistance against Bt toxin when they feed on the GM corn plants.
D. corn borers which are resistant against Bt toxin would pass on the relevant genes to the next generation.

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMA0430000023|!| ★★

The following diagram shows the production of a type of pluripotent stem cells (PSCs).



Source: Y Tambe /CC BY-SA 3.0, text labels added to original file

Which of the following is an advantage of using this type of PSCs over embryonic stem cells?

A. It avoids the destruction of embryos.
B. It avoids the formation of tumours.
C. It uses adult skin cells, which can differentiate into more cell types.
D. It uses a vector, which can introduce genes into cells more efficiently.

##A##

Skill assessed	Mark
Applying concepts	1

© 2024 Aristo Educational Press Ltd.

|!|BCEMB0430000005|!| ★

Which of the following are social concerns related to growing GM crops?

(1) GM crops may monopolize the market and consumers may not have non-GM crops to choose from.
(2) Biotechnology companies own the rights to sell expensive seeds for GM crops.
(3) GM crops may out-compete natural wild type plants and disrupt the ecosystem.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

□

##A##

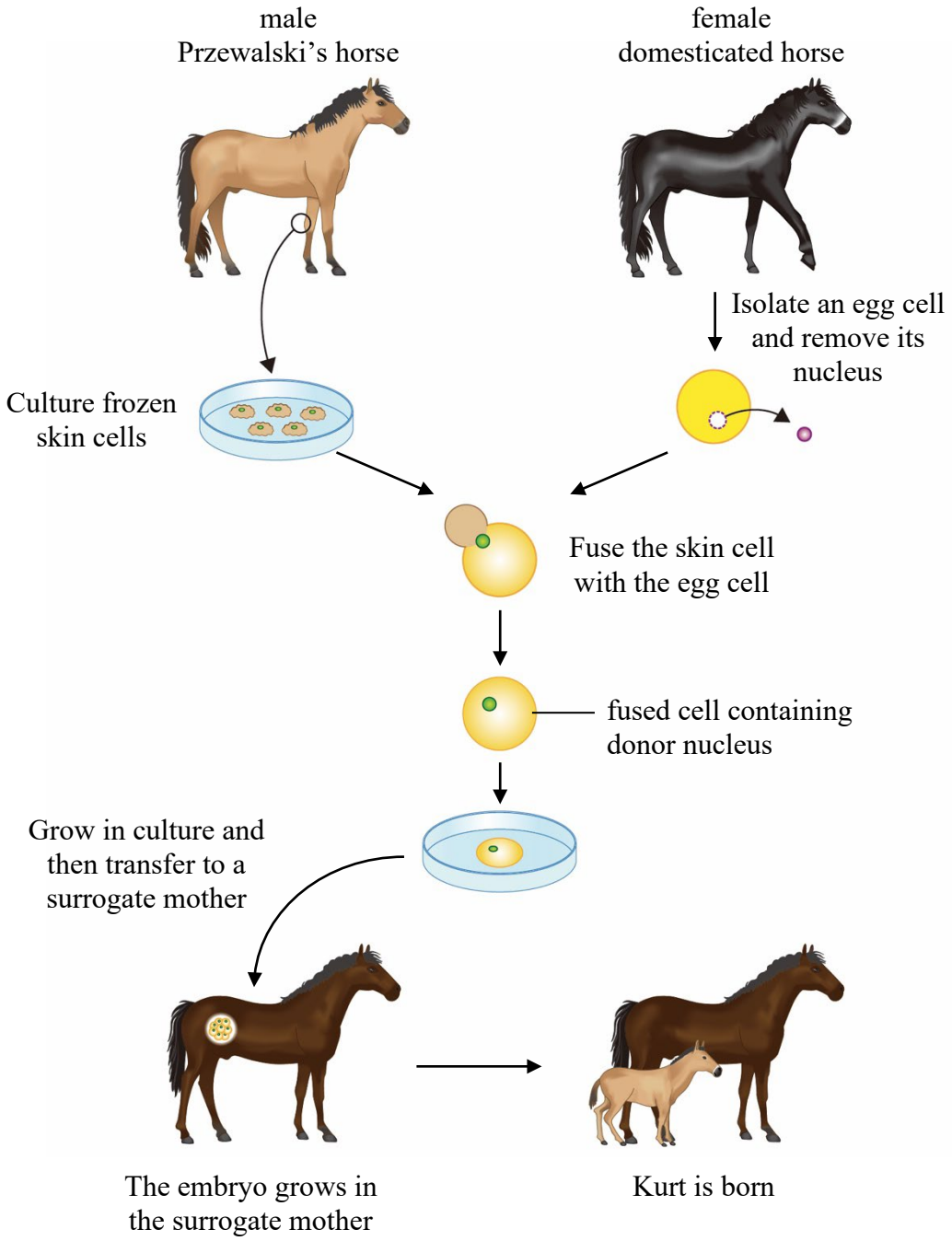
Skill assessed	Mark
Applying concepts	1

© 2024 Aristo Educational Press Ltd.

B. Structured questions

|!|BCELB0410000021|!| ★ **Cross topic**

The Przewalski's horse is an endangered species that has been extinct in the wild and only lived in zoos. Scientists have successfully cloned Kurt, a male Przewalski's horse from DNA that has been preserved for more than 40 years in frozen environment. The figure below shows the cloning process of Kurt:



(a) Name this method of cloning. (1 mark)

(b) (i) Which horse is Kurt genetically identical to? (1 mark)

(ii) Explain why Kurt's height may be different from that of the horse in (i). (2 marks)

Most of the surviving Przewalski's horses today live in zoos, and have been bred from only 12 wild horses. Scientists hope that Kurt can help bring genetic diversity to the diminishing population of the Przewalski's horse.

(c) (i) What problems may result from breeding the surviving Przewalski's horses with each other? Explain your answer. (2 marks)

(ii) Explain how Kurt can bring genetic diversity to the surviving population of Przewalski's horse. (3 marks)

##

(a) Somatic cell nuclear transfer (1)

(b) (i) The male Przewalski's horse (1)

(ii) The same genes in Kurt that control body height may be expressed differently, compared with the male Przewalski's horse. (1)
Kurt may have a different diet compared with the male Przewalski's horse. (1)
(Or other reasonable answers)

(c) (i) The surviving horses are bred from the same 12 horses and all have similar DNA base sequences. (1)
Inbreeding between these horses may result in genetic diseases / defects. (1)

© 2024 Aristo Educational Press Ltd.

(ii) Kurt's DNA comes from a source stored over 40 years ago and is significantly different from the surviving horses in zoos today. (1)
Kurt can breed naturally with the surviving female Przewalski's horses. (1)
Independent assortment of homologous chromosomes and random fusion of gametes will result in genetically diverse offspring. (1)

(9 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	1
(b)	Analysing data	1
(b)(ii)	Applying concepts	2
(c)(i)	Applying concepts	2
(c)(ii)	Applying concepts	3

|!|BCELB0430000012|!| ★ **Cross topic**

A person likes his plants and pet dog very much, and is thinking about cloning them. He is discussing his idea with a scientist.

(a) (i) State a method to create genetically identical plants without using modern biotechnology. (1 mark)

(ii) Describe a suitable cloning method if the person would like to make cloned plants free of pathogens. (4 marks)

(b) Concerning the process of animal cloning using nuclear transfer, explain why cloning the person's pet dog is ethically more controversial than cloning plants. (2 marks)

© 2024 Aristo Educational Press Ltd.

(c) The scientist suggests another method. In this method, the person's pet dog breeds with another dog with similar desirable traits that the person likes.

(i) In terms of the means by which new lives are formed, explain why this method is less controversial than animal cloning. (2 marks)

(ii) Widespread practice of the method of breeding animals with similar traits may give rise to inbreeding. State how inbreeding and animal cloning can lead to health problems in the animals reproduced respectively. (2 marks)

##

(a) (i) Artificial propagation (1)

(ii) Obtain sterilized explants from the parent plant. (1)
Culture the explant in a sterile culture medium with plant hormones and essential nutrients, forming a callus. (1)
Culture pieces of callus in another sterile culture medium containing plant hormones, forming plantlets. (1)
Plant the plantlets formed in the soil. (1)

(b) Cloning animals often involves the destruction of unsuccessfully cloned embryos, while cloning plants do not involve the formation of embryos. (1)
Cloning animals raises the ethical problem of sacrificing embryos for only a small number of successful ones. (1)

(c) (i) Artificial breeding still depends on the natural process of breeding between dogs. (1)
However, animal cloning artificially gives rise to new lives, instead of using natural biological mechanisms of reproduction. (1)

(ii) Cloned animals may develop abnormally, such as having abnormally large organs. (1)
Inbreeding easily gives rise to offspring with health defects related to recessive alleles. (1)

(11 marks)

© 2024 Aristo Educational Press Ltd.

##

	Skills assessed	Marks
(a)(i)	Understanding basic concepts	1
(a)(ii)	Applying concepts	4
(b)	Applying concepts	2
(c)(i)	Applying concepts	2
(c)(ii)	Applying concepts	2

|!|BCELB0420000008|!| ★ **STSE**

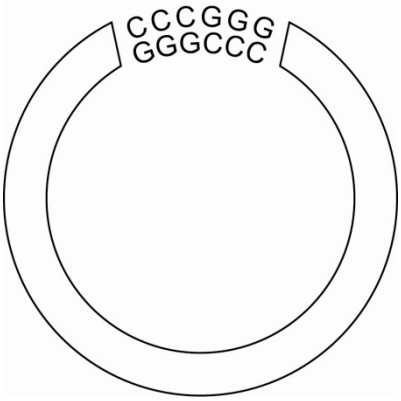
Restriction enzymes recognize specific base sequences and cut DNA molecules at specific restriction sites.

The restriction sites of restriction enzymes *Hpa*II, *Bsh*I, *Sma*I and *Xma*I are shown in the table below. The arrows show where the enzymes cut the DNA molecule.

Restriction enzyme	Restriction site
*Hpa*II	↓ – C – C – G – G – – G – G – C – C – ↑
*Bsh*I	↓ – G – G – C – C – – C – C – G – G – ↑
*Sma*I	↓ – C – C – C – G – G –G – – G – G – G – C – C –C – ↑
*Xma*I	↓ – C – C – C – G – G – G – – G – G – G – C – C – C – ↑

© 2024 Aristo Educational Press Ltd.

(a) The following figure shows a plasmid which contains the restriction sites of the restriction enzymes *Sma*I and *Xma*I.



State whether the base sequence of this plasmid can also be cut by the restriction enzymes *Hpa*II and *Bsh*I. (2 marks)

(b) A scientist cuts a plasmid with *Xma*I in order to insert a human gene.

(i) Describe the steps for inserting the human gene into the plasmid with the use of *Xma*I. (3 marks)

(ii) If the plasmid was cut with *Sma*I, suggest what might happen during the procedure of human gene insertion. (2 marks)

© 2024 Aristo Educational Press Ltd.

(c) Patients with coronary heart disease can be treated by gene therapy. In gene therapy, restriction enzyme is used to cut out a human gene Z. This gene codes for a protein growth factor which stimulates the growth of blood vessels. The isolated gene Z is enveloped within a liposome and injected into the heart to promote the growth of blood vessels.

(i) Suggest another vector for the transfer of human gene to the heart. (1 mark)

_____

(ii) What is the advantage of using liposome over the vector in (c)(i)? (1 mark)

_____

(iii) Suggest ***one*** benefit and ***one*** potential hazard of the gene therapy mentioned above. (2 marks)

_____

_____

##

(a) *Hpa*II can cut this base sequence. (1)
*Bsh*I cannot cut this base sequence. (1)

(b) (i) The plasmid is cut by *Xma*I to expose its sticky ends. (1)
The DNA fragments containing the human gene must be cut with *Xma*I. (1)
The complementary sticky ends of the cut human gene and that of the cut plasmid can join together by DNA ligase. (1)

(ii) *Sma*I cuts the plasmid and produces 'blunt ends'. (1)
There is no complementary sequence to guide the joining of DNA, thus the human gene may be inserted in the wrong direction, or the plasmid may rejoin without a human gene insert. (1)

(c) (i) virus / retrovirus / other reasonable answer (1)

(ii) It is a safer method as it does not involve any microorganisms. (1)

© 2024 Aristo Educational Press Ltd.

(iii) Any ***one*** of the following benefits: (1)

- New blood vessels grow and supply nutrients and oxygen to the heart muscle.
- New blood vessels grow and carry metabolic wastes away from the heart muscle.

(or other reasonable answers)

Any ***one*** of the following hazards: (1)

- The expression of another gene may be disrupted.
- The new genes may be wrongly delivered or be inappropriately expressed.

(or other reasonable answers)

(11 marks)

##

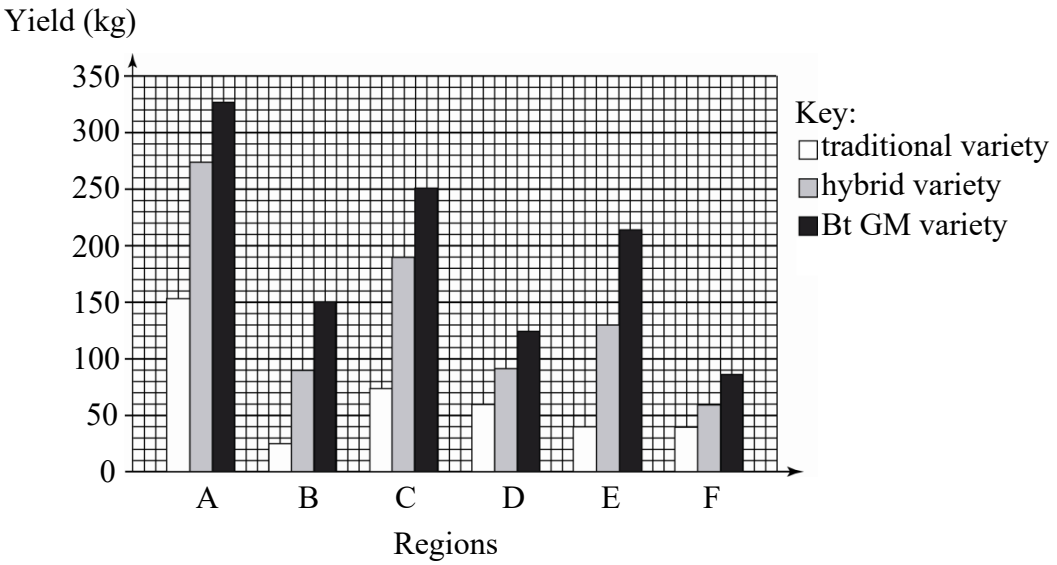
	Skills assessed	Marks
(a)	Analysing data	2
(b)(i)	Applying concepts	3
(b)(ii)	Applying concepts	2
(c)(i)	Understanding basic concepts	1
(c)(ii)	Applying concepts	1
(c)(iii)	Critical thinking	2

© 2024 Aristo Educational Press Ltd.

|!|BCELB0420000018|!| ★

Traditional varieties of maize are easily damaged by insect pests. New varieties of maize, which include hybrid varieties and GM varieties, have been developed to reduce the loss due to insect pests.

In an investigation, scientists studied the yields of different varieties of maize crops in six regions of southern Africa. The following graph shows the results.



(a) Briefly describe how a hybrid variety of maize can be produced. (3 marks)

(b) Explain why it is necessary to use the polymerase chain reaction (PCR) during the production of the Bt GM variety of maize. (2 marks)

(c) According to the graph, what conclusions can be drawn from the results? (2 marks)

© 2024 Aristo Educational Press Ltd.

(d) In spite of the above results, some farmers still prefer the hybrid variety to the Bt GM variety. Suggest ***two*** reasons to explain why. (2 marks)

_____

_____

_____

##

(a) Crossbreed two maize plants with desirable characteristics. (1)
Select only the offspring with the desirable characteristics and cultivate them. (1)
Repeat the selective breeding process until all offspring have the desirable characteristics. (1)

(b) The quantity of DNA with the Bt toxin gene obtained from the bacteria (Bt) is very small. (1)
PCR is used to amplify the Bt toxin gene. (1)

(c) The Bt GM variety has the highest yield, followed by the hybrid variety, while the traditional variety has the lowest yield. (1)
In general, the yields of all varieties of maize are the lowest in region F and highest in region A. (1)
(or other reasonable answers)

(d) Any ***two*** of the following: (2)
- The production cost of Bt GM seeds is higher than that of hybrid seeds.
- The hybrid variety may have other good characteristics (e.g. better taste / drought resistance etc.)
- Bt GM maize may cause allergy in some people / have unknown long-term effects on health.

(or other reasonable answers)

(9 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	3
(b)	Applying concepts	2
(c)	Analysing data	2
(d)	Applying concepts	2

© 2024 Aristo Educational Press Ltd.